

September 3, 2026

**Prof. Roland Rathelot**

Editor

*The Economic Journal*

**Subject:** Revision of “Boots in the Beat: Effects of Beat Policing on Victimization and Crime Perception”

Dear Prof. Rathelot,

Thank you for giving us the opportunity to revise our manuscript further. We are delighted that Referee 3 has signed off on the paper, and we carefully read your comments and the comments from Referees 1 and 2. We are very grateful for this detailed and constructive feedback, which helped us further improve the paper.

Below we provide a point-by-point response document with detailed answers to each comment. We also indicate how the manuscript has been revised in response.

Sincerely,

Andrés Barrios Fernández on behalf of the team

## **Editor's comments**

### **Editor Comment 1**

Dear Doctor Barrios-Fernandez,

Thank you for submitting your fascinating paper to The Economic Journal.

I have now heard back from the reviewers and read your revised paper. The news is rather good. R3 has signed off on the paper. R2 has a few comments, which are rather conditional-acceptance material. R1 is pleased with the revision but still has issues regarding the interpretation of your results and the overall framing of the paper. I agree with them. I think we have established that your paper makes a contribution to the literature; now is the time to acknowledge very precisely what can and cannot be said about the results. Please provide further evidence to support the current interpretation of the results if you can and if the evidence supports it (the reviewer has suggestions). Also, please revise the framing so that the results, interpretations, and framing are well aligned. At this stage, I would like you to be mindful of readers and start condensing your paper to maximise readability and impact. I think there is room to cut around 5 pages and to make the introduction (and the abstract) shorter, in particular (of course, do not cut anything you find crucial).

You will find detailed comments in R1's and R2's reports. As usual, I would like you to prepare a detailed, separate reply for each reviewer and revise the paper accordingly. Additionally, please prepare a separate note for me, outlining the changes made during the revision process.

### **Response to Editor Comment 1:**

## **References**

## Referee 1 Suggestions, Comments and Questions

Thank you very much for your careful reading of the revised manuscript and for finding the causal evidence on the effects of *Plan Cuadrante* convincing. We have carefully considered your remaining comments on interpretation and framing and hope that we adequately addressed them in the revised manuscript and in the response document below.

### Comment R1.1

**1. What does Plan Cuadrante identify?** The paper states in various places that permanent geographic assignment and greater local knowledge are the key mechanisms underlying the RF effects of the policy. I think this interpretation is stronger than what the evidence can establish.

Plan Cuadrante appears to have changed several features of frontline policing simultaneously. Assigning officers to smaller quadrants changes not only the officer-area relationships, but also the spatial organization of police presence. Depending on quadrant size, number of officers, and deployment practices, the reform may have effectively changed police density and patrol intensity, as well as response times. Each of these margins could plausibly affect crime, victimization, perceptions of safety, and citizen cooperation. Greater local knowledge is one compelling mechanism within this bundle, but it is not separately observed from these other operational changes.

I therefore think the paper should explicitly acknowledge that the results identify the effect of a broader organizational and deployment reform rather than the effect of local knowledge alone. A more defensible interpretation would be that Plan Cuadrante, combining permanent officer-to-quadrant allocation with an increase in police density, improved public safety and citizens' perceptions of the police. Even holding the total number of officers fixed, reorganizing officers across smaller territorial units may change the intensity and concentration of local police presence.

One potentially useful exercise would be to characterize police density across quadrants, for example in terms of officers per square kilometer (or per capita). The illustrative map in the paper suggests that quadrants vary substantially in geographic size (and the smaller quadrants are the ones with higher baseline crime rates). If the authors observe the number of officers assigned to each quadrant, it would therefore be useful to document the distribution of officers per quadrant, and examine whether treatment effects vary with this measure of effective local police density. Police density may itself be correlated with response times, patrol intensity, and other operational margins, so it would still not isolate a single mechanism. But it would help characterize an important dimension of what the reform changed.

Response to R1.1:

### Comment R1.2

**2. Resource changes.** Figure 4 is most informative in the comparison between the bottom 40 percent of treated municipalities and control municipalities, which experienced similar resource growth. The interpretation should nevertheless be cautious. In particular, not even the bottom group or the control group experience zero resource growth. The exercise therefore does not isolate the effects of the geographic reorganization, absent those of changes in resources. Rather, it shows that the estimated gains are not confined to municipalities experiencing the largest resource expansions.

Relatedly, I think Table B3 is helpful. I would not dismiss these specifications simply because resource changes are themselves induced by treatment. Conditioning on policy-induced resource changes can still be informative about whether the reduced-form effect is concentrated in places experiencing large resource expansions.

## **Response to R1.2:**

### **Comment R1.3**

Minor point: in Panel D in Table B3, the sample size falls substantially when the officer control is added. Why?

## **Response to R1.3:**

Thanks for the opportunity to clarify this point. This is related with the unit of analysis. The outcome analyzed in Panel D is the number of crimes reported to the police per 100,000 inhabitants, which is available at the municipality-year level. This is administrative data provided by the sub-Secretary for Crime Prevention. In contrast, the outcomes reported in panels A to C are measured in the ENUSC survey, which allows the unit of analysis to be at the individual level (these are repeated cross-sections), although the treatment variable is defined at the municipality-year level also in this analysis. Note also the survey did not include geocodes or more spatially granular information.

### **Comment R1.4**

**3. Contribution and framing.** I would simplify the framing of the paper's contribution. The findings provide evidence that the organization and territorial deployment of police resources matter for public safety (and for citizens' perceptions, which by itself is a quite novel result), even though the specific mechanisms generating these gains cannot be fully decomposed. I think this is an interesting and important contribution on its own, and it does not require attributing the effects specifically to local knowledge.

## **Response to R1.4:**

### **Comment R1.5**

By contrast, I do not find the analogy in the introduction to the spatial organization of health and education services particularly convincing. The underlying economic problem is different. Health and education services are typically supplied at fixed locations, with citizens incurring costs to access them, and supply being constrained. Here the relevant margin is about how a mobile public service is deployed across space. I understand the broad conceptual similarity the authors have in mind, but I do not think the analogy is sufficiently close to motivate a distinct contribution to the public/org. econ. literature along this dimension. I would therefore suggest dropping or substantially de-emphasizing this part of the introduction.

**Response to R1.5:**

Thanks for the suggestion. We have removed the paragraph discussing this analogy from the introduction.

**References**

## Referee 2 Suggestions, Comments and Questions

Thank you very much for your careful reading of the revised manuscript and response, and for confirming that it reads much better as a result. We have carefully considered all your remaining comments, in particular your suggestions on the presentation of the spillover evidence, and hope that we adequately addressed them in the revised manuscript and in the response document below.

### Comment R2.1

Regarding my main concern—spillovers—the current version is considerably more transparent, and Section 4.2, together with the Online Appendix, attempts to address this threat to identification. I would, however, encourage the authors to reorganize the presentation of this evidence. My suggestion would be to lead with the exposure-based specification, in which exposure to treated neighbors is measured continuously. In my view, this is the most informative of the three exercises because it exploits variation in the intensity of potential spillovers rather than relying on a binary indicator for having a treated neighbor. This specification should, of course, be presented with the appropriate caveat: exposure to treated neighbors is not randomly assigned, so the estimates do not fully address the possibility that municipalities with many early-adopting neighbors differ systematically in other respects.

### Response to R2.1:

### Comment R2.2

I also had difficulty following the randomization-inference exercise for spillovers. This may simply reflect my own misunderstanding, but although I understand randomization inference in general, the purpose of this particular exercise as a test for spillovers was unclear to me. It may therefore warrant further explanation. In particular, how does the exercise help address concerns about spillovers? Why is treatment status reshuffled across all 346 municipalities when significantly fewer municipalities were ever eligible for the program?

### Response to R2.2:

Thanks for pointing this out, and for pushing us to explain the purpose of this exercise more clearly.

We should first flag an important difference with the exercise’s inspiration, Blattman et al. (2021): in their setting, treatment is randomly assigned by the researchers, whereas in ours, adoption of *Plan Cuadrante* is not random. This matters, because it means our exercise cannot be “randomization inference” in the strict Fisherian sense of permuting according to a known, true random-assignment mechanism to obtain an exact-size test. What we do instead is closer to a placebo/permutation-based benchmarking exercise: we ask what our estimates would typically look like if adoption dates and never-treated status carried no real information—i.e., if they were assigned as if by chance—and compare our actual estimates to that benchmark distribution.

This is precisely how the exercise speaks to spillovers, which was your specific question. For each of 1,000 random reassignments of adoption dates and never-treated status, we re-estimate

both (i) the direct effect on treated municipalities and (ii) the spillover effect on their untreated neighbors, building a placebo distribution for each. We then locate our two actual point estimates within their respective placebo distributions. The direct effect falls in the bottom first percentile of its placebo distribution—it is far more negative than almost anything a chance reassignment of adoption dates would generate, which is what we would expect if *Plan Cuadrante* has a real, sizeable effect on treated municipalities. The spillover effect, by contrast, falls near the center of its placebo distribution (between the 40th and 45th percentile)—indistinguishable from the kind of small estimate that chance reassignment routinely produces. This is exactly the pattern we would expect if there is no genuine spillover: a placebo world with no true spillovers should produce estimates that look just like the ones we actually obtain, which is what we find. In other words, the exercise complements our other two spillover tests by showing that our near-zero spillover estimate is not an outlier relative to a “no information” benchmark, the same way the direct effect is shown to be an outlier relative to that same benchmark.

The Blattman et al. (2021) connection is nonetheless useful for a second, related reason, which we should also make explicit: they show that when treatment or spillover status is assigned by proximity to other treated units rather than by a clean partition of the sample, nearby units end up sharing the same status purely by chance. This generates what they call “fuzzy clustering”—implicit clusters that do not correspond to any observed grouping variable, vary in size, and can be correlated with potential outcomes—which can bias standard cluster-robust inference, and even point estimates, if uncorrected. Our own spillover specifications assign treatment and neighbor-spillover status at the municipality level, which is also our clustering unit, so municipality boundaries (unlike distance-based spillover bands) give us a clean partition of the sample; we therefore do not think our own standard errors are exposed to the same fuzzy-clustering bias. But since our spillover exercise borrows their permutation approach, we thought it useful to explain this feature of their design as well.

Finally, in response to your specific question about the reshuffling universe: rather than reassigning adoption dates and never-treated status across the full national sample of 346 municipalities—most of which were below the 70% urban-population threshold and were therefore never eligible for *Plan Cuadrante*, and so could not have adopted it under any circumstance—we now restrict the placebo reassignment to the 150 municipalities that met the eligibility criterion. This keeps the placebo/benchmark exercise anchored to the same population that could, in principle, have received the treatment, instead of mixing in municipalities that were never at risk of adoption.

### **Comment R2.3**

I found a few small inconsistencies in the response to my report—for example, references to Figure 4 where Figure 6 appears to have been intended—but I checked these against the revised manuscript and did not find the same inconsistencies there. There is one exception. In response to one of my comments, the authors quote revised text for Online Appendix C.1 stating that “the effects on arrests are not statistically significant in the ENUSC subsample.” This sentence does not appear in the manuscript, which still reads only: “While the effects for reported crime are larger in magnitude, the estimates are qualitatively similar...”

### **Response to R2.3:**

#### **Comment R2.4**

I also noticed that some estimates based on the administrative data have changed between the two versions. In particular, the estimated reduction in reported crime is now 10 percent rather than 14 percent. I could not find this change documented in the response. As long as the authors can provide a reasonable explanation for what changed between rounds and are confident that the current result is robust, I do not view this as a major concern.

#### **Response to R2.4:**

Thanks for pointing this out, and apologies for not documenting this change explicitly in our previous response.

The change in the headline number—from a 14% to a 10% decline in reported crime—does not come from a change in the underlying treatment effect: the  $\beta$  coefficient from the Callaway and Sant’Anna estimator is essentially unchanged between rounds. It comes instead from a change in how we compute  $\hat{Y}$ , the baseline level we use as the denominator to convert that level effect into a percentage change. In the previous version,  $\hat{Y}$  was computed by pooling the pre-treatment mean among treated municipalities with the mean among never-treated municipalities. Since the Callaway and Sant’Anna estimator is essentially a treatment-on-the-treated (ToT) parameter, we now believe it is more appropriate to compute  $\hat{Y}$  using only the pre-treatment observations of treated municipalities. This revision affects the administrative-data-based estimates much more than the ENUSC-based ones, because the administrative sample includes 196 never-treated municipalities, whereas the ENUSC sample includes only one; pooling into  $\hat{Y}$  therefore moved the administrative baseline by much more than the ENUSC one. We apologize again for not flagging this change explicitly in our earlier response. To avoid this ambiguity going forward, we now define  $\hat{Y}$  explicitly in the notes of every figure that reports it.

A second, unrelated and more minor change may also explain small differences in some  $\beta$  coefficients across rounds, though not in the result you highlight. Between the first and second submission we realized that our implementation of the Callaway and Sant’Anna estimator was not weighting estimates by cohort size, while the standard implementation does. We corrected this, which slightly changes some point estimates but not in a way that is economically or statistically meaningful.

#### **Comment R2.5**

Finally, I noted a few minor editorial issues:

1. Two spaces are missing: on p. 7, “full municipal territory.On average,” and on p. 21, “improve security perceptions,police forces need.”

#### **Response to R2.5:**

Thanks for the catch. Both missing spaces are now corrected.



## Comment R2.6

2. The abbreviation "PCSP" does not appear to be defined in the paper, although it is used in some notes and figures.

### Response to R2.6:

Thanks for the catch. We now introduce the program's full name, Plan Cuadrante de Seguridad Pública, the first time it is mentioned in the institutional background section (page XX), noting that for simplicity we refer to it as Plan Cuadrante throughout. We also replaced the remaining instances of "PCSP" in figure notes with "PC" for consistency; the acronym "PCSP" now only appears in the bibliography.

## Comment R2.7

3. The reference list includes Ajzenman, Dominguez, and Undurraga (2023a) and (2023b) as separate entries for what appears to be the same paper, and both entries are cited in the text.

### Response to R2.7:

Thanks for the catch. This is now corrected: we removed the duplicate entry and all in-text citations now point to the single remaining reference.

## References

Blattman, C., D. P. Green, D. Ortega, and S. Tobón (2021, 02). Place-Based Interventions at Scale: The Direct and Spillover Effects of Policing and City Services on Crime. *Journal of the European Economic Association* 19(4), 2022–2051.